Classical-quantum gravity as quantum gravity in disguise



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Paper + Collaborators

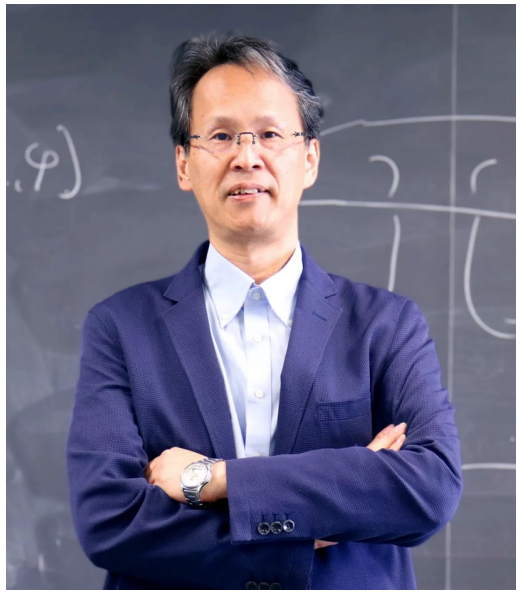
Classical-quantum gravity as quantum gravity in disguise

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Do we need to quantise gravity?

PHYSICAL REVIEW X **13,** 041040 (2023)

Featured in Physics

A Postquantum Theory of Classical Gravity?

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Any consistent coupling between classical gravity and quantum matter is fundamentally irreversible

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nature communications

Article

Gravitationally induced de... space-time diffusion: testing the quantum nature of gravity

Received: 12 July 2023

Accepted: 8 November 2023

Jonathan Oppenheim[1] ✉, Carlo Sparaciari[1], Barbara Šoda[1,2,3] & Zachary Weller-Davies[1,3]

White paper: 1-10 Hz matter-wave interferometer to test the spin entanglement witness for quantum gravity

REVIEWS OF MODERN PHYSICS, VOLUME 97, JANUARY–MARCH 2025

Quantum-information methods for quantum gravity laboratory-based tests

Chiara Marletto and Vlatko Vedral

Clarendon Laboratory, University of Oxford, Parks Road, Oxford OX1 3PU, United Kingdom

(published 14 March 2025)

The question is no longer philosophical. *Now:* **model-building + phenomenology**

Do we need to quantise gravity?



Jonathan Oppenheim
@postquantum

21 January 2021
Twitter · 1/8

I don't know whether space-time geometry is fundamentally quantum or classical, but @carlorovelli has given me 5000:1 odds, so I'm taking him up on it. When he looses the bet, he has promised to give me 5000 of some item, but what item should I choose? youtu.be/yeXdSSMfY3U?t=2139 1/



Jonathan Oppenheim
@postquantum

21 January 2021
Twitter · 3/8

He wins the bet if an experiment such as this one: arxiv.org/abs/1707.06050 demonstrates the quantum nature of gravity, while I win the bet when an experiment shows that gravity is fundamentally classical (a spatially covariant theory was proposed in arxiv.org/abs/1811.03116). 3/

A postquantum theory of classical gravity?
The effort to discover a quantum theory of gravity is motivated by the need to recon...
https://arxiv.org/abs/1811.03116

A Spin Entanglement Witness for Quantum Gravity
Understanding gravity in the framework of quantum mechanics is one of the great c...
https://arxiv.org/abs/1707.06050

Outline

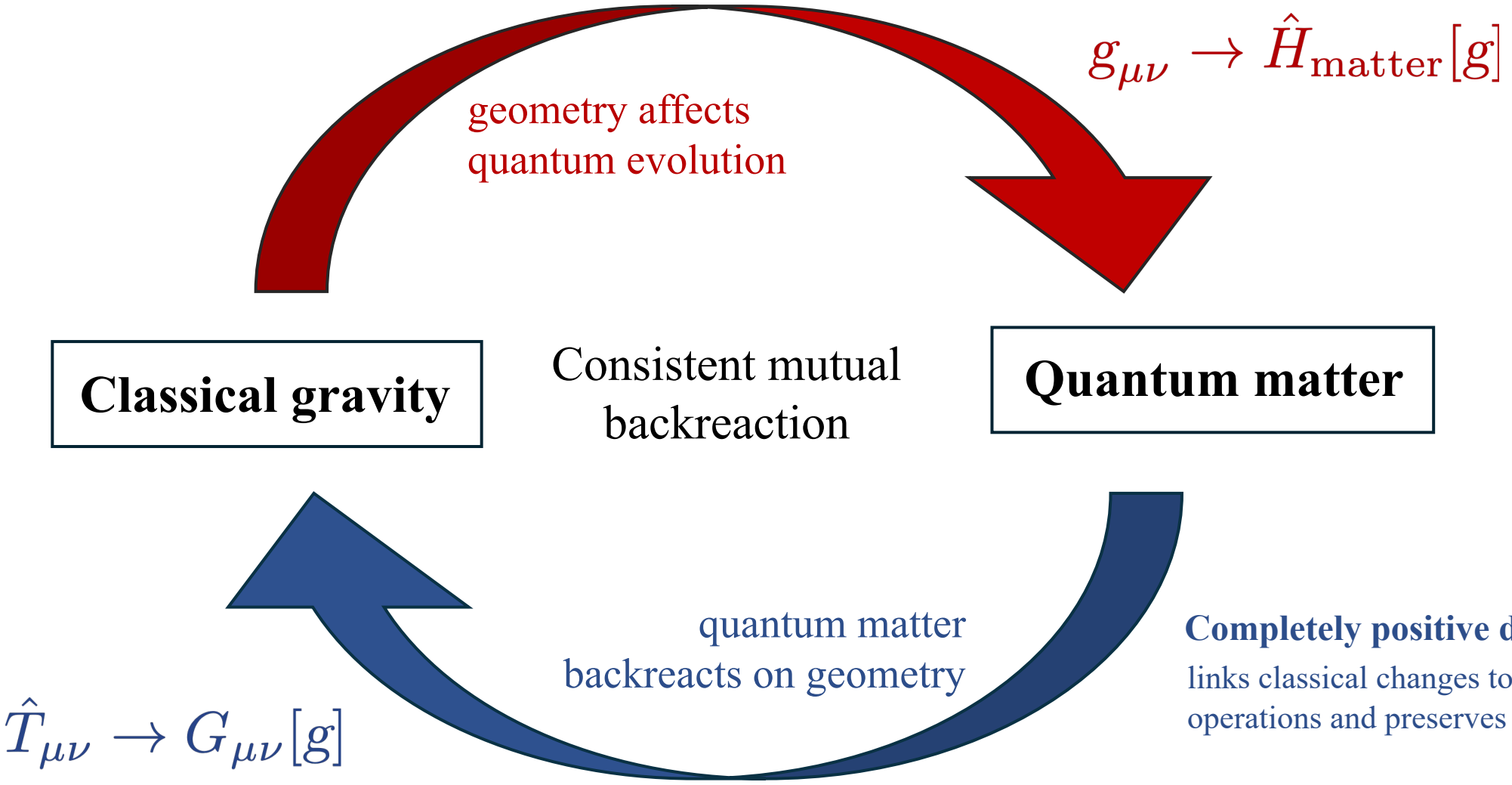
Main question: Can completely positive classical-quantum theories be fundamental?

1. What is a classical-quantum (CQ) state?
2. Completely positive (CP) dynamics
3. **Embedding CQ theories into a quantum theory**

Main result of our paper

4. Toy model: violation of angular momentum conservation

Classical-quantum gravity



Completely positive dynamics
links classical changes to quantum operations and preserves positivity

Attitudes toward hybrid dynamics

Hybrid as approximation

- ❑ Valid for limited times
- ❑ Breakdown signals validity limits
- ❑ Classical sector is effective
- ❑ Used for computation

Hybrid as fundamental theory

- ❑ No expiration date
- ❑ Breakdown is a consistency problem
- ❑ Classical sector is fundamental
- ❑ Used as ontology

What is a classical-quantum state?

CQ system: density operator $\hat{\rho}(z,t)$
that depends on classical variables $z = (q,p)$

Classical system: $\varrho(z,t) = \mathrm{Tr}_Q[\hat{\rho}(z,t)]$

Quantum system: $\rho_Q(t) = \int \hat{\rho}(z,t)\, dz$

Normalization:

$$\int \mathrm{Tr}_Q[\hat{\rho}(z,t)]\, dz = 1$$

Why completely positive dynamics?

Reversible linear CQ schemes: conceptually attractive, but generally inconsistent

CPCQ schemes: formally consistent, but irreversible

Markovian evolution $\Lambda_t = e^{t\mathcal{L}}$

valid states/probabilities ➡ CPTP maps ➡ GKSL-type dynamics

positivity + normalization + arbitrary ancillas

$$(\mathrm{id}_R \otimes \Lambda_Q)[\rho_{RQ}] \geqslant 0$$
$$\forall\, R, \rho_{RQ} \geqslant 0$$

familiar from open quantum systems

Can the CP hybrid formalism be truly fundamental?

The CQ master equation

Terms local in z (act at a fixed classical configuration)

$$\frac{\partial \hat{\rho}(z,t)}{\partial t} = \overbrace{-i[H(z),\hat{\rho}(z,t)]}^{\textbf{quantum evolution}} + \overbrace{\mathcal{D}_{\rm loc}[\hat{\rho}(z,t)]}^{\textbf{local noise}}$$

$$+ \underbrace{\int W(z|z')[\hat{\rho}(z',t)]\,dz' - \tfrac{1}{2}[R(z),\hat{\rho}(z,t)]_+}_{\textbf{CQ transitions}}$$

incoming gain (nonlocal) — outgoing loss (local)

$$R(z) := \sum_{\mu,\nu} \int W^{\mu\nu}(z'|z) L_\nu^\dagger L_\mu \, dz',$$

$$[A,B]_+ := AB + BA.$$

Statistical moments

Previous slide: $W(z|z')$: classical jump $z' \to z$ plus quantum operation on Q.

Idea: describe the $z' \to z$ transition by moments of $\Delta z = z - z'$.

Fokker-Planck equation

Kramers-Moyal expansion

$$D_{(n)}^{\mu\nu\, i_1 \cdots i_n}(z') := \frac{1}{n!} \int H^{\mu\nu}(z|z') \prod_{k=1}^{n} \left(z^{i_k} - z'^{\,i_k} \right) dz$$

$(\Delta z)^n$ is encoded in $D_{(n)}$

$$\partial_t \hat{\rho} = \underbrace{\mathcal{L}_{\rm cl}\hat{\rho}}_{\text{purely classical}} + \underbrace{\mathcal{L}_Q(z)\hat{\rho}}_{\text{quantum at fixed } z} + \underbrace{\mathcal{L}_{\rm br}\hat{\rho}}_{\text{backreaction}}$$

with $$\mathcal{L}_{\rm br}\hat{\rho} = \sum_{\substack{(\mu,\nu) \\ \neq(0,0)}} \sum_{n \geq 1} (-1)^n \partial_{i_1} \cdots \partial_{i_n} \left[D_{(n)}^{\mu\nu i_1 \cdots i_n}(z) L_\mu \hat{\rho} L_\nu^\dagger \right].$$

Backreaction is constrained by CP

$$\mathcal{L}_{\rm br}\hat{\rho} = \sum_{\substack{(\mu,\nu)\\ \neq(0,0)}} \sum_{n\geq 1} (-1)^n \partial_{i_1}\cdots\partial_{i_n}\left[D^{\mu\nu i_1\cdots i_n}_{(n)}(z) L_\mu \hat{\rho} L^\dagger_\nu\right].$$

Physical significance:

$D^{(0)} \sim$ decoherence/dissipation scale

$$n=1: \quad -\partial_i\left(D^{(1)i}_{\rm br}\hat{\rho}\right) \rightsquigarrow \text{drift/force on } z$$

$$n=2: \quad \partial_i\partial_j\left(D^{(2)ij}_{\rm br}\hat{\rho}\right) \rightsquigarrow \text{diffusion/noise in phase space}$$

Constrained by complete positivity:

$$2\left\langle D^{(2)}_{\rm br}\right\rangle\left\langle D^{(0)}\right\rangle \succeq \left\langle D^{(1)}_{\rm br}\right\rangle\left\langle D^{(1)}_{\rm br}\right\rangle^\dagger$$



Oppenheim *et al.*
Nat. Commun. **14**, 7910 (2023)

CP backreaction requires decoherence and diffusion.

Embedding theorem

Is CPCQ dynamics genuinely hybrid, or can it be represented inside a quantum theory?

Step 1: Encode each possible classical value z as a basis state of a quantum register Z.

$$z \mapsto |z\rangle_Z, \quad \Pi_z = |z\rangle_Z\langle z|_Z$$

Step 2: Represent the CQ state as a block-diagonal state on QZ.

$$\rho_{QZ}(t) = \sum_z \hat{\rho}_Q(z,t) \otimes \Pi_z$$

Step 3: Recover the CQ evolution from a CPTP map on QZ.

$$\langle z|\Gamma_t[\rho_{QZ}(0)]|z\rangle = \hat{\rho}_Q(z,t)$$

Why the diagonal sector? Classical observables on Z are diagonal:

$$A_Z = \sum_z a_z \Pi_z$$

Theorem:

For discrete or coarse-grained z,
CPCQ evolution is the Z-diagonal sector of a CPTP map on QZ.

The CQ transition rule

We work with a discrete classical label z.

$$\hat{\rho}_Q(z,t) = \sum_{z',\alpha} K_{\alpha Q}(z|z',t)\, \hat{\rho}_Q(z',0)\, K^{\dagger}_{\alpha Q}(z|z',t)$$

$$z' \xrightarrow{K_{\alpha Q}(z|z',t)} z$$

transition + operation on Q

Kraus operators: $K_{\alpha Q}(z|z',t) : \; Q \mapsto K_{\alpha Q}(z|z',t)(\,\cdot\,)K^{\dagger}_{\alpha Q}(z|z',t)$

Completeness: $\sum_{\alpha,z} K^{\dagger}_{\alpha Q}(z|z',t)K_{\alpha Q}(z|z'',t) = \delta_{z'z''} I_Q$

Trace preservation: $\sum_z \mathrm{Tr}_Q[\hat{\rho}_Q(z,t)] = \sum_{z'} \mathrm{Tr}_Q[\hat{\rho}_Q(z',0)]$

Goal: CQ transition rule ⟹ quantum channel on QZ

Construct the QZ channel

Previous slide: $K_{\alpha Q}(z|z',t)$ combines two pieces: 1. change of classical label $z' \to z$
2. operation on Q.

Once z is stored in Z, the label change $z' \to z$ becomes a register update $|z'\rangle_Z \to |z\rangle_Z$.

$$\Xi_\alpha(t)_{QZ} = \sum_{z,z'} K_{\alpha Q}(z|z',t) \otimes |z\rangle_Z\langle z'|_Z$$

$$|\psi\rangle_Q \otimes |z'\rangle_Z \xrightarrow{\Xi_\alpha} \sum_z K_{\alpha Q}(z|z',t)|\psi\rangle_Q \otimes |z\rangle_Z$$

acts on Q updates the register Z: $z' \to z$

Quantum channel: $\Gamma_t[\rho_{QZ}] = \sum_\alpha \Xi_\alpha(t)_{QZ}\, \rho_{QZ}\, \Xi_\alpha^\dagger(t)_{QZ}$

$$\sum_\alpha \Xi_\alpha^\dagger(t)\Xi_\alpha(t) = I_{QZ}$$

(inherited from CQ completeness)

CQ transition rule ⟹ CPTP map on QZ

Recover the CQ state

We now check that the quantum channel on QZ reproduces the original CQ evolution.

Initial embedded state: $\rho_{QZ}(0) = \sum_{z'} \hat{\rho}_Q(z',0) \otimes \Pi_{z'}$ $\qquad \langle z'|\rho_{QZ}(0)|z''\rangle = \delta_{z'z''}\hat{\rho}_Q(z',0)$

$$\langle z|\Gamma_t[\rho_{QZ}(0)]|z\rangle = \sum_{\alpha,z'} K_{\alpha Q}(z|z',t)\,\hat{\rho}_Q(z',0)\,K^{\dagger}_{\alpha Q}(z|z',t) = \hat{\rho}_Q(z,t)$$

➡ Exactly the CQ transition rule

Pinching (keep only Z-diagonal blocks):

$$\mathcal{P}_Z[X] = \sum_z (I_Q \otimes \Pi_z) X (I_Q \otimes \Pi_z)$$

$$\rho_{QZ}(t) = \mathcal{P}_Z(\Gamma_t[\rho_{QZ}(0)]) = \sum_z \hat{\rho}_Q(z,t) \otimes \Pi_z$$

QZ channel ➡ CQ dynamics on the Z-diagonal blocks

off-diagonal Z blocks

Before pinching:

$$\Gamma_t[\rho_{QZ}(0)] = \begin{pmatrix} \hat{\rho}_Q(z_1,t) & * & * \\ * & \hat{\rho}_Q(z_2,t) & * \\ * & * & \hat{\rho}_Q(z_3,t) \end{pmatrix}_Z$$

After pinching: CQ state

$$\mathcal{P}_Z(\Gamma_t[\rho_{QZ}(0)]) = \begin{pmatrix} \hat{\rho}_Q(z_1,t) & 0 & 0 \\ 0 & \hat{\rho}_Q(z_2,t) & 0 \\ 0 & 0 & \hat{\rho}_Q(z_3,t) \end{pmatrix}_Z$$

Operational classicality

Previous slide: the Z-diagonal blocks reproduce the CQ state.

$$A_Z = \sum_z a_z \Pi_z, \quad \Pi_z = |z\rangle_Z\langle z|_Z$$

$$A_Z, B_Z \in \mathcal{A}_Z \implies [A_Z, B_Z] = 0$$

classical observables = commuting functions of z

Why can we ignore the off-diagonal blocks?

Off-diagonal blocks are invisible: $\mathrm{Tr}[(A_Q \otimes A_Z)X] = \mathrm{Tr}[(A_Q \otimes A_Z)\mathcal{P}_Z[X]]$

$$\mathrm{Tr}\big[(A_Q \otimes A_Z)\big(X - \mathcal{P}_Z[X]\big)\big] = 0$$

not observable by Z measurements

$$X = \begin{pmatrix} X_{11} & * & * \\ * & X_{22} & * \\ * & * & X_{33} \end{pmatrix}_Z \xrightarrow{\text{diagonal } Z\text{-readout}} \{X_{11}, X_{22}, X_{33}\}$$

The accessible Z algebra is classical.

Quantum, but not canonical quantization

What kind of "quantum theory" did we obtain?

Canonical quantization

$$z = (h, \pi)$$
$$h, \pi \longrightarrow \hat{h}, \hat{\pi} \qquad [\hat{h}, \hat{\pi}] \sim i\hbar$$

Promotes phase-space variables to noncommuting operators.

Our embedding

$$z = (h, \pi) \longrightarrow |z\rangle_Z \qquad \Pi_z = |z\rangle_Z \langle z|_Z$$

$$\hat{h}_Z = \sum_z h(z)\Pi_z, \quad \hat{\pi}_Z = \sum_z \pi(z)\Pi_z \qquad [\hat{h}_Z, \hat{\pi}_Z] = 0$$

The register stores $z = (h, \pi)$ as a label.
h and π are diagonal functions of that label.

Z provides a commutative readout algebra, not canonical gravitational degrees of freedom.

$$\Gamma_t \text{ is CPTP on } QZ \Longrightarrow \text{unitary completion on } QZE$$

$$\Gamma_t[\rho_{QZ}] = \mathrm{Tr}_E \left[U_{QZE}(t) \left(\rho_{QZ} \otimes |0\rangle_E \langle 0|_E \right) U^\dagger_{QZE}(t) \right]$$

Implications for CQ gravity

The embedding result shows that CPCQ gravity need not be irreducibly hybrid.

$$\rho_{QZ}^{\mathrm{CQ}}(t) = \mathcal{P}_Z\left(\Gamma_t[\rho_{QZ}^{\mathrm{CQ}}(0)]\right)$$

Γ_t: CPTP map on QZ, $\mathcal{P}_Z$: diagonal Z-readout

fully quantum QZE $\xrightarrow{\mathrm{Tr}_E}$ CPTP channel on QZ $\xrightarrow{\mathcal{P}_Z,\ \mathcal{A}_Z}$ CPCQ dynamics

What the embedding shows:

CPCQ dynamics admits an exact realisation as a restricted sector of a fully quantum theory.

Open questions:

1. What is actually realised in nature?
2. Are Noether charges conserved in the CQ sector?

Does the reduced CP sector inherit Noether conservation laws?

Symmetry vs. conservation in CP dynamics

In Hamiltonian/unitary dynamics, symmetries and conserved generators are tightly linked.

In CP dynamics, they are separate conditions.

Closed Hamiltonian dynamics

$$\dot{\rho} = -i[H, \rho] \qquad \frac{d}{dt}\langle A \rangle = i\langle [H, A] \rangle$$

$$\boxed{[A, H] = 0 \implies A \text{ conserved}}$$

CP/Lindblad-type dynamics

$$\dot{\rho} = \mathcal{L}[\rho] \qquad \frac{d}{dt}\langle A \rangle = \langle \mathcal{L}^\dagger(A) \rangle$$

$$\boxed{\frac{d}{dt}\langle A \rangle = 0 \ \ \forall\, \rho \iff \mathcal{L}^\dagger(A) = 0}$$

$$\boxed{\text{covariance of } \mathcal{L} \not\Longrightarrow \mathcal{L}^\dagger(G) = 0}$$

Symmetry does not guarantee conservation of its generator G.

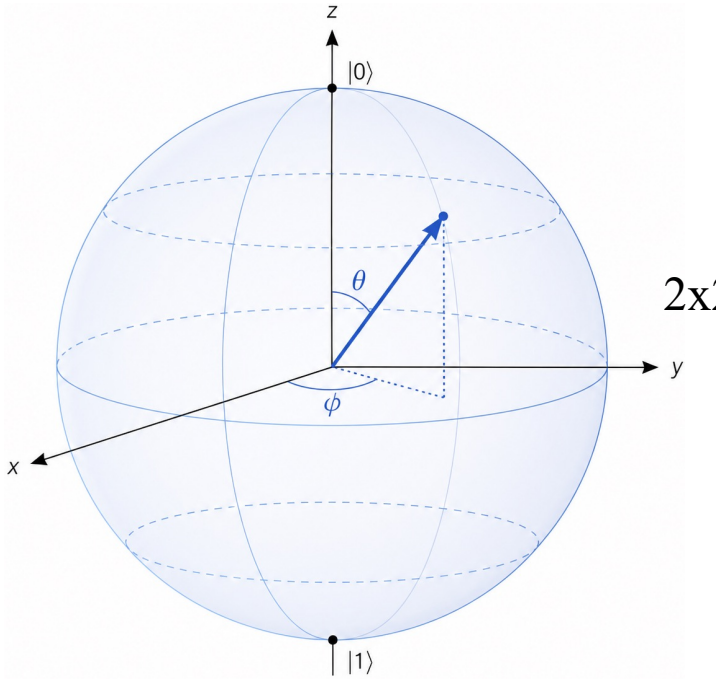
Toy model: qubit + classical particle

Minimal closed CQ model with rotational symmetry.

Quantum subsystem Q

$$\hat{\sigma}_a, \quad a \in \{x, y, z\}$$

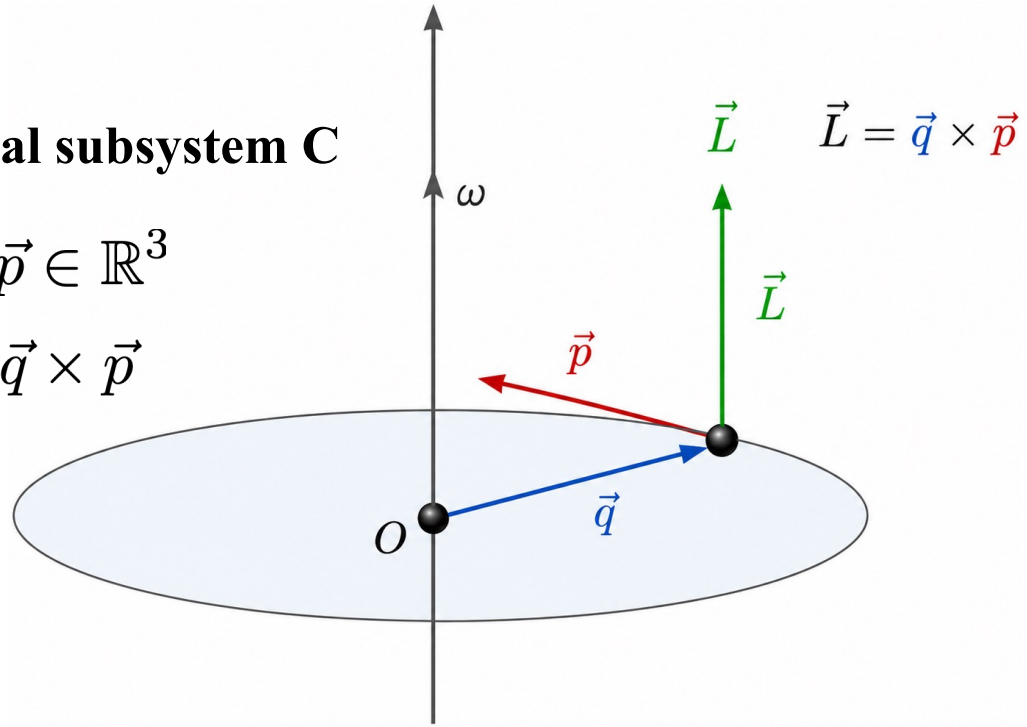
$$\hat{S}_a = \frac{\hbar}{2}\hat{\sigma}_a$$



Classical subsystem C

$$\vec{q}, \vec{p} \in \mathbb{R}^3$$

$$\vec{L} = \vec{q} \times \vec{p}$$



CQ system

$$\hat{\rho}(\vec{q}, \vec{p}, t)$$

2x2 density matrix depending on classical phase space variables

Total angular momentum: $\hat{J}_a(\vec{q}, \vec{p}) = L_a(\vec{q}, \vec{p})\, I_2 + \frac{\hbar}{2}\hat{\sigma}_a$

Is $\langle \hat{J}_a \rangle$ conserved if the CQ dynamics is rotationally invariant?

Rotationally invariant CQ dynamics

Now: choose CPCQ evolution that has no preferred spatial direction.

relaxation/decoherence rate

rotationally invariant distribution

$$\frac{\partial}{\partial t}\hat{\rho}(\vec{q},\vec{p},t) = \underbrace{-\kappa\,\hat{\rho}(\vec{q},\vec{p},t)}_{\text{relaxation of current CQ state}} + \underbrace{\frac{\kappa}{4}\,\rho_f(|\vec{q}|,|\vec{p}|,\vec{q}\cdot\vec{p})}_{\text{repopulation into rotationally invariant state}} \int \left[\hat{\rho}(\vec{q}\,',\vec{p}\,',t) + \sum_{a=x,y,z} \hat{\sigma}_a \hat{\rho}(\vec{q}\,',\vec{p}\,',t)\hat{\sigma}_a\right] d^3q'\, d^3p'$$

no preferred spin axis

Stationary state: $\hat{\rho}_\infty(\vec{q},\vec{p}) = \frac{I_2}{2}\,\rho_f(|\vec{q}|,|\vec{p}|,\vec{q}\cdot\vec{p})$

maximally mixed qubit + rotationally invariant classical distribution

The dynamics is rotationally invariant, but not Hamiltonian/unitary.

Solution: relaxation to an isotropic state

The equation can be solved exactly.

$$\hat{\rho}(\vec{q},\vec{p},t) = e^{-\kappa t}\hat{\rho}(\vec{q},\vec{p},0) + \frac{I_2}{2}\left(1-e^{-\kappa t}\right)\rho_f(|\vec{q}|,|\vec{p}|,\vec{q}\cdot\vec{p})$$

initial CQ state survives with weight $e^{-\kappa t}$

relaxation to rotationally invariant stationary state

Long-time limit: $t \to \infty$: $\hat{\rho}(\vec{q},\vec{p},t) \longrightarrow \frac{I_2}{2}\rho_f(|\vec{q}|,|\vec{p}|,\vec{q}\cdot\vec{p})$

maximally mixed spin + rotationally invariant classical distribution

All directional information in the initial state is suppressed by $e^{-\kappa t}$.

Rotational symmetry, but angular momentum decays

Choose a rotationally invariant stationary distribution with $\vec{L}_f = 0$: $\rho_f(\vec{q},\vec{p}) = \delta^3(\vec{q})\,\delta^3(\vec{p})$

Take a localized initial state: $\hat{\rho}(\vec{q},\vec{p},0) = \delta^3(\vec{q}-\vec{q}_0)\delta^3(\vec{p}-\vec{p}_0)\hat{\rho}_i$

Initial total angular momentum: $\langle \hat{J}_a(0)\rangle = (\vec{q}_0 \times \vec{p}_0)_a + \frac{\hbar}{2}\,\mathrm{Tr}[\hat{\sigma}_a\hat{\rho}_i]$ $\qquad$ $\langle \hat{J}_a(t)\rangle = \langle L_a(t)\rangle + \frac{\hbar}{2}\langle \hat{\sigma}_a(t)\rangle$

$$\frac{d}{dt}\langle L_a\rangle = -\kappa\langle L_a\rangle \qquad \frac{d}{dt}\langle \hat{\sigma}_a\rangle = -\kappa\langle \hat{\sigma}_a\rangle$$

$$\langle L_a(t)\rangle = e^{-\kappa t}(\vec{q}_0 \times \vec{p}_0)_a \qquad \frac{\hbar}{2}\langle \hat{\sigma}_a(t)\rangle = \frac{\hbar}{2}e^{-\kappa t}\,\mathrm{Tr}[\hat{\sigma}_a\hat{\rho}_i]$$

$$\boxed{\langle \hat{J}_a(t)\rangle = e^{-\kappa t}\langle \hat{J}_a(0)\rangle}$$

$$\boxed{\langle \hat{J}_a(\infty)\rangle = 0}$$

$$\text{rotational symmetry} \not\Longrightarrow \frac{d}{dt}\langle \hat{J}_a\rangle = 0$$

Rotationally invariant CPCQ dynamics does not imply angular-momentum conservation.

Lessons from the toy model

The toy model highlights a structural issue of CPCQ dynamics.

$$\text{symmetry of reduced CP equations} \not\Rightarrow \text{conservation of the associated charge}$$

What we showed: Closed CQ system + CP dynamics + rotational symmetry $\langle \hat{J}_a(t) \rangle = e^{-\kappa t} \langle \hat{J}_a(0) \rangle$

Important caveat: Toy model $\neq$ full relativistic CQ gravity

Phenomenological constraints: $\kappa \lesssim \frac{1}{T}$ $2 \langle D_{\rm br}^{(2)} \rangle \langle D^{(0)} \rangle \succeq \langle D_{\rm br}^{(1)} \rangle \langle D_{\rm br}^{(1)} \rangle^{\dagger}$

Bounds on relaxation or nonconservation constrain the allowed CP backreaction/diffusion terms.

Quantum embedding does not automatically protect conservation laws in the reduced CQ sector.

Take-home message

1. CPCQ dynamics can be embedded into fully quantum dynamics.
2. The classical variables are read out from a commuting Z register.
3. The reduced sector can violate conservation laws despite symmetries.

CQ gravity may therefore be quantum gravity with restricted access.

Classical-quantum gravity as quantum gravity in disguise

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Backup slides

Decoherence, diffusion, and all that

Decoherence: suppresses quantum coherences

Diffusion: spreads a distribution in phase space

Dissipation: relaxes or drains conserved quantities such as energy

Stochasticity: the underlying randomness that can produce these effects

The CQ master equation

$$\frac{\partial \hat{\rho}(z,t)}{\partial t} = -i[H(z,t), \hat{\rho}(z,t)] + \lambda^{\mu\nu}(z,t)\left(L_\mu \hat{\rho}(z,t) L_\nu^\dagger - \tfrac{1}{2}[L_\nu^\dagger L_\mu, \hat{\rho}(z,t)]_+\right)$$
$$+ \int \left(W^{\mu\nu}(z|z',t) L_\mu \hat{\rho}(z',t) L_\nu^\dagger - \tfrac{1}{2} W^{\mu\nu}(z'|z,t)[L_\nu^\dagger L_\mu, \hat{\rho}(z,t)]_+\right) dz'. \qquad (2.5)$$

Physical form: separates Hamiltonian evolution, local Lindblad noise, and nonlocal CQ transitions.

$$\frac{\partial \hat{\rho}(z,t)}{\partial t} = \int H^{\mu\nu}(z|z') L_\mu \hat{\rho}(z',t) L_\nu^\dagger \, dz'$$
$$- \tfrac{1}{2} H^{\mu\nu}(z)[L_\nu^\dagger L_\mu, \hat{\rho}(z,t)]_+. \qquad (2.9)$$

Kernel form: packages the same generator into $H^{\mu\nu}(z|z')$ whose jump moments define $D_{(n)}$.

What does this imply for CQ gravity?

The toy model highlights a structural issue of CPCQ dynamics.

$$\text{symmetry of reduced CP equations} \not\Longrightarrow \text{conservation of the associated charge}$$

What we showed

1. Closed CQ system.
2. CP dynamics.
3. Rotationally invariant EOM.
4. But: $\langle \hat{J}_a(t) \rangle = e^{-\kappa t} \langle \hat{J}_a(0) \rangle$

Symmetry survives; conservation fails.

What this does not yet show

1. Toy model is not full theory.
2. Relativistic CPCQ gravity must be checked separately.
3. Constraints may depend on specific gravitational implementation.

Toy model $\neq$ full relativistic CQ gravity

Phenomenological constraints

no observed relaxation over time T

bounds on $\kappa \implies$ bounds on $\langle D^{(0)} \rangle$, $\langle D^{(1)}_{\text{br}} \rangle$, $\langle D^{(2)}_{\text{br}} \rangle$

Quantum embedding $\not\Longrightarrow$ conservation laws in the reduced CQ sector

Recover the CQ state

We now check that the quantum channel on QZ reproduces the original CQ evolution.

Initial embedded state: $\rho_{QZ}(0) = \sum_{z'} \hat{\rho}_Q(z',0) \otimes \Pi_{z'}$ $\qquad \langle z'|\rho_{QZ}(0)|z''\rangle = \delta_{z'z''}\hat{\rho}_Q(z',0)$

$$\begin{aligned}\langle z|\Gamma_t[\rho_{QZ}(0)]|z\rangle &= \sum_{\alpha,z',z''} K_{\alpha Q}(z|z',t)\langle z'|\rho_{QZ}(0)|z''\rangle K^\dagger_{\alpha Q}(z|z'',t) \\ &= \sum_{\alpha,z'} K_{\alpha Q}(z|z',t)\,\hat{\rho}_Q(z',0)\,K^\dagger_{\alpha Q}(z|z',t) \\ &= \hat{\rho}_Q(z,t)\end{aligned}$$

➡ Exactly the CQ transition rule

Before pinching:

$$\Gamma_t[\rho_{QZ}(0)] = \begin{pmatrix} \hat{\rho}_Q(z_1,t) & * & * \\ * & \hat{\rho}_Q(z_2,t) & * \\ * & * & \hat{\rho}_Q(z_3,t) \end{pmatrix}_Z$$

After pinching:

$$\mathcal{P}_Z(\Gamma_t[\rho_{QZ}(0)]) = \begin{pmatrix} \hat{\rho}_Q(z_1,t) & 0 & 0 \\ 0 & \hat{\rho}_Q(z_2,t) & 0 \\ 0 & 0 & \hat{\rho}_Q(z_3,t) \end{pmatrix}_Z$$

Diagonal projection: $\mathcal{P}_Z[X] = \sum_z (I_Q \otimes \Pi_z) X (I_Q \otimes \Pi_z)$

$$\rho_{QZ}(t) = \mathcal{P}_Z(\Gamma_t[\rho_{QZ}(0)]) = \sum_z \hat{\rho}_Q(z,t) \otimes \Pi_z$$

QZ channel ➡ CQ dynamics on the Z-diagonal blocks